

MATH 345 Skeleton Notes

Unit 2: Anatomy of a linear map

Section 2.2: Linear systems

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The preview examples had different stories: reachable outputs, invisible changes, redundant features. Algebraically, they all ask the same question: can a list of linear equations in the unknown input coordinates be solved?

Definitions and examples

Definition: Linear equations

An equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where $a_1, \dots, a_n, b$ are scalars, and $x_1, \dots, x_n$ are variables is called

a

. The real quantities $a_1, \dots, a_n$ are called

, b is called a

, while the variables

$x_1, \dots, x_n$ are called *unknowns*.

Definition: Systems of linear equations

A

or

is a set of m linear equations each in n

unknowns. A linear system has the form

$$\begin{array}{ccccccc}
a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\
a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\
\vdots & & \vdots & & & & \vdots & & \vdots \\
a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m
\end{array}$$

where $a_{11}, \dots, a_{mn}$ and $b_1, \dots, b_m$ are constants.

A to the linear system the referenced item is a sequence of n numbers $s_1, \dots, s_n$ so that *each* equation in the referenced item is satisfied when the substitutions $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ are made.

Warning: Variable names

We use x, y, z in geometric examples and $x_1, \dots, x_n$ for general systems.

Activity: Nonlinear equations (U2-LO1)

What are some equations that are not linear equations?

Before introducing a general algorithm, we look at a few small systems where the equations can be analyzed directly.

Activity: Forest animals (U2-LO1, U2-LO2)

In a Wisconsin forest, there are robins and badgers. Together they have 18 heads and 56 legs. How many robins and badgers are in the forest?

Activity: Identifying coefficients, constants, and unknowns (U2-LO1)

Identify the coefficients, the constants and the unknowns in each linear equation in Forest animals (U2-LO1, U2-LO2).

Definition: Consistency and equivalence

- If a linear system has *no* solution, it is called .
- If a linear system has at least one solution, it is called . Note that an equation with infinitely many solutions is still consistent!
- If two linear systems have exactly the same solutions, they are called .

Activity: Yet another forest problem (U2-LO1, U2-LO2)

In another Wisconsin forest, there are deer and badgers. Together they have 18 heads and 56 legs. How many deer and badgers are in the forest? Is the resulting linear system consistent or inconsistent?

Geometric examples in $\mathbb{R}^2$ and $\mathbb{R}^3$: building intuition

Recall that a line in the plane which passes through a point $P_0(x_0, y_0)$, and has slope m , can be written in *point-slope form*, as a set of points $P(x, y)$ which solve the equation

$$y - y_0 = m(x - x_0).$$

Example: The geometric point of view (U2-LO1, U2-LO2)

Consider the following three systems, inspired by our forest examples. Let's graph the lines defined by these equations and check how this reflects the number of solutions in each case.

Task

System 1

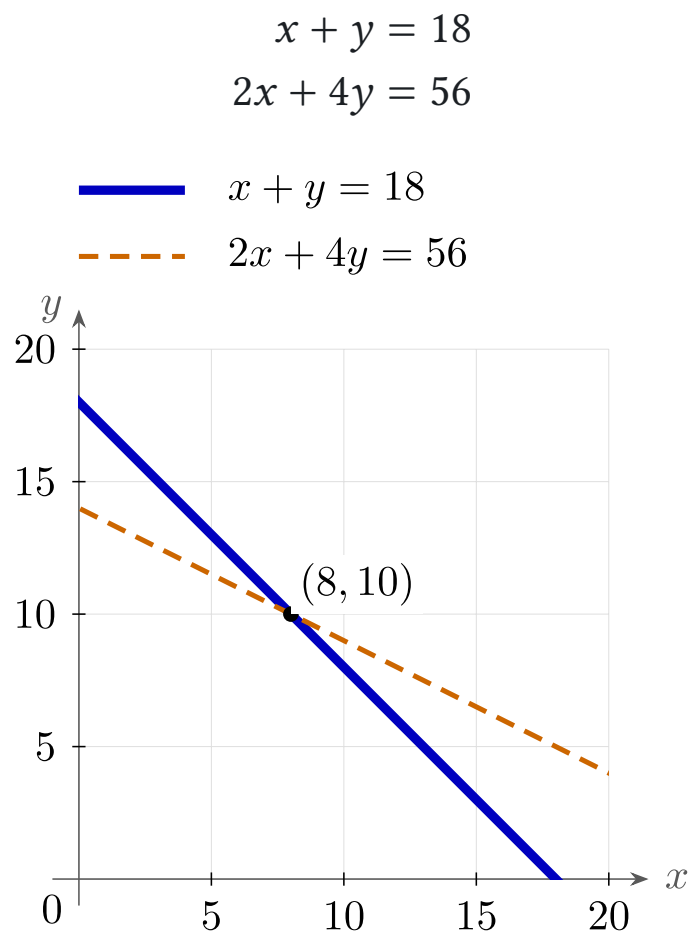


Figure. System 1: the two lines intersect at (8, 10).

Task

System 2

$$\begin{aligned}x + y &= 18 \\4x + 4y &= 56\end{aligned}$$

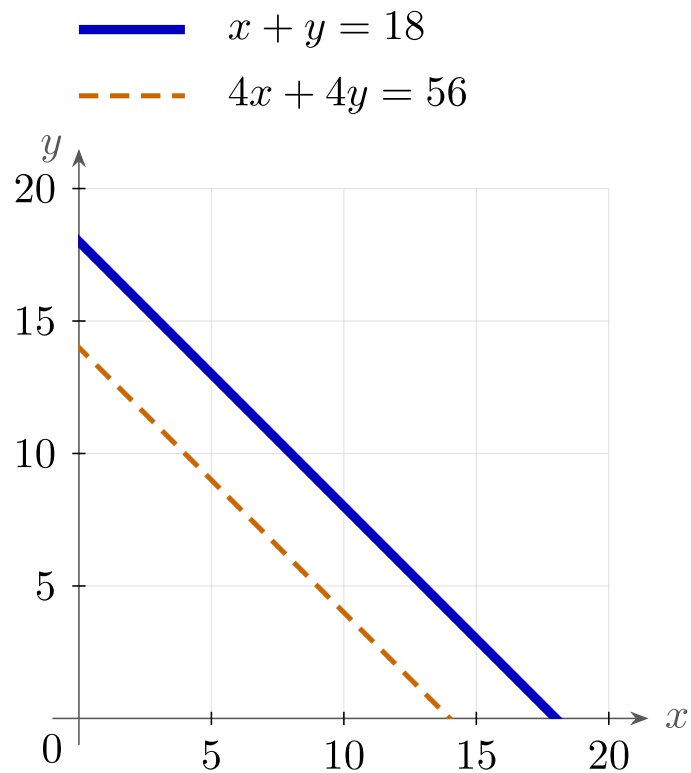


Figure. System 2: the two lines are parallel and distinct, so there is no solution.

Task

System 3

$$\begin{aligned}x + y &= 18 \\4x + 4y &= 72\end{aligned}$$

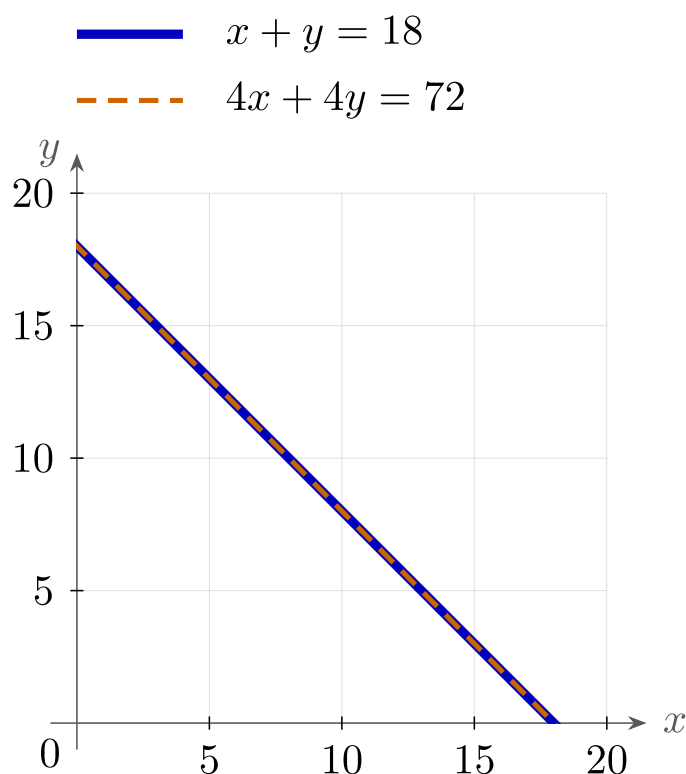


Figure. System 3: the two lines coincide, giving infinitely many solutions.

System 1 has one intersection point: subtracting $x + y = 18$ from $x + 2y = 28$ gives $y = 10$, so $x = 8$. System 2 gives $x + y = 18$ and $x + y = 14$, so the two lines are parallel and the system has no solution. System 3 is the same line twice, since $4x + 4y = 72$ is equivalent to $x + y = 18$, so it has infinitely many solutions.

The same word “solution” also has a geometric meaning. A solution set can be a point, a line, a plane, empty, or higher-dimensional. In three dimensions, the solution set of a linear equation is a plane, provided at least one variable has a nonzero coefficient. To obtain an equation for such a plane, we take a different approach from the point-slope form used for lines in two dimensions. We use a normal vector, which points at right angles to the plane and thus gives a direction in which the plane *does not travel*.

Definition: Normal vector equation for planes

The plane passing through $P_0(x_0, y_0, z_0)$ with $\mathbf{n}$

is the set of all points $P(x, y, z)$ such that $\mathbf{x} - \mathbf{x}_0$ is orthogonal to $\mathbf{n}$. That is,

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{x}_0) = 0,$$

or in scalar form, if $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$ and $\mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix}$,

$$n_1(x - x_0) + n_2(y - y_0) + n_3(z - z_0) = 0.$$

In general form, this equation is written as

$$ax + by + cz = d,$$

where $a = n_1$, $b = n_2$, $c = n_3$, and

$$d = n_1x_0 + n_2y_0 + n_3z_0.$$

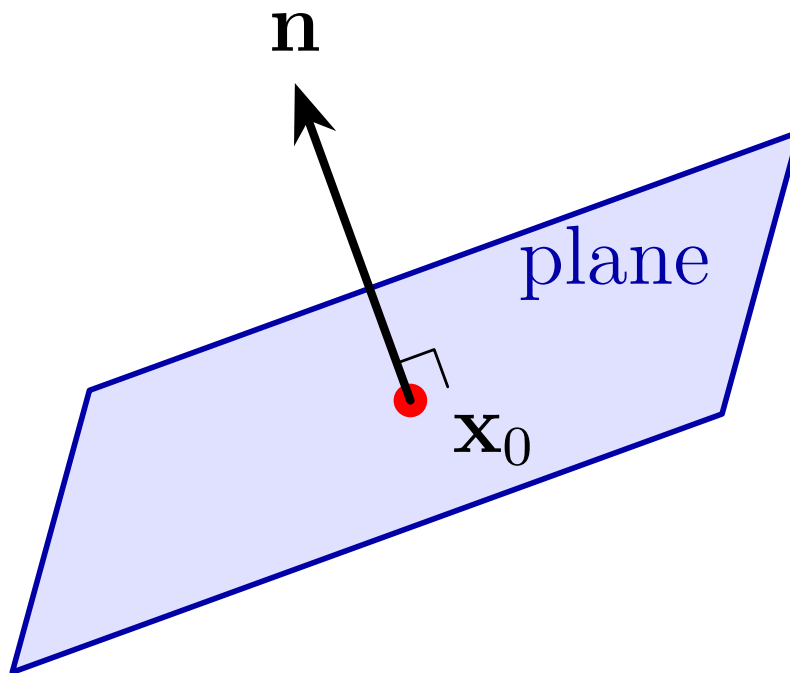


Figure. A plane passing through a point $\mathbf{x}_0$ with normal vector $\mathbf{n}$.

Activity: A plane from a point and a normal vector (U2-LO1)

Find the equation in general form of the plane passing through the point $P(1, 0, 3)$, normal to the vector $\mathbf{n} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$.

Shapes in three-dimensional space described as the zeros of a single equation are typically *two-dimensional*. Lines are described as points which satisfy *two* different equations. If a line in three-dimensional space is equal to the intersection of two planes passing through a point $\mathbf{x}_0$ (see the figure), with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, then the line is precisely the set of points $\mathbf{x}$ which satisfy the two equations

$$\mathbf{n}_1 \cdot (\mathbf{x} - \mathbf{x}_0) = 0$$

and

$$\mathbf{n}_2 \cdot (\mathbf{x} - \mathbf{x}_0) = 0.$$

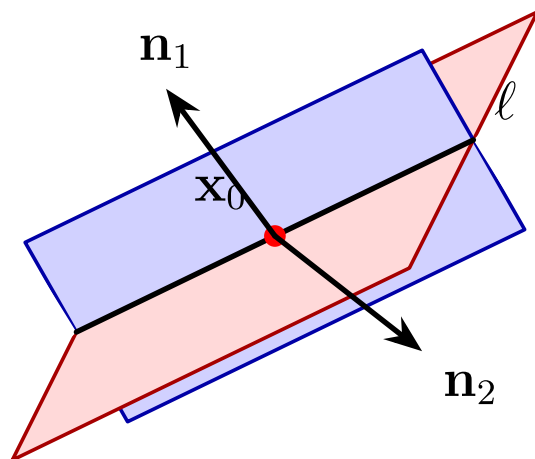


Figure. A line defined as the intersection of two planes, with normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, passing through a common point $\mathbf{x}_0$.

Definition: Parametric equation for a line

The line through $P_0(x_0, y_0, z_0)$ with $\mathbf{d}$ is the set of points $P(x, y, z)$ of the form

$$\mathbf{x} = \mathbf{x}_0 + t\mathbf{d},$$

where $\mathbf{x}_0$ is the vector corresponding to P_0 . In coordinates, if $\mathbf{d} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$,

this says

$$(x, y, z) = (x_0 + td_1, y_0 + td_2, z_0 + td_3).$$

If d_1, d_2, d_3 are all nonzero, we can also write the line *in symmetric form* as the set of points $P(x, y, z)$ such that

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$$

All three values in this equation are equal to the scalar t which occurs in the first equations.

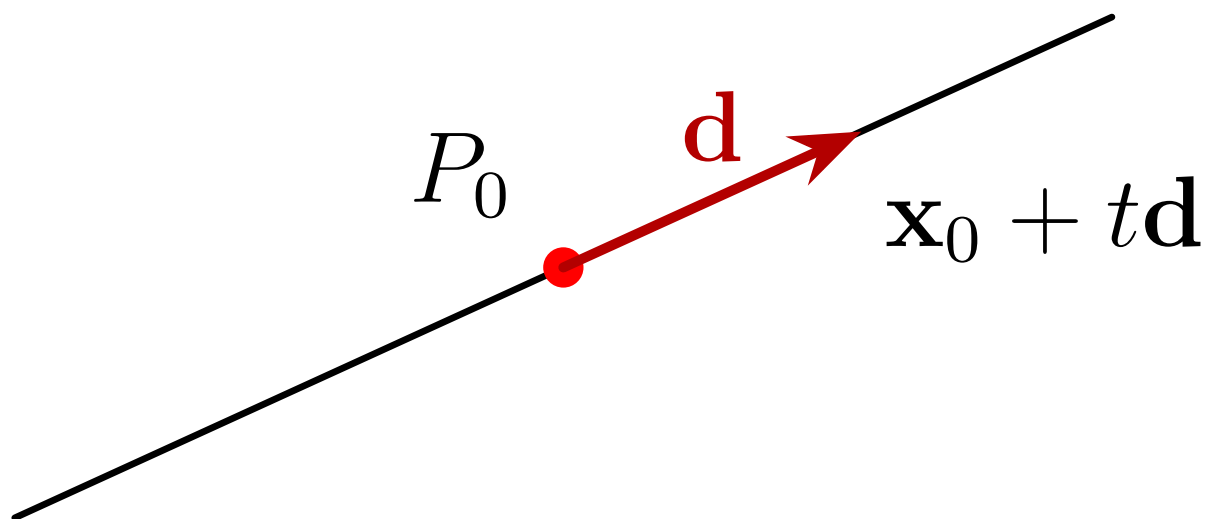


Figure. A line defined by a direction vector $\mathbf{d}$ and a point P_0 , with corresponding vector $\mathbf{x}_0$.

Activity: A parametric line through two points (U2-LO1)

Find a parametric equation for the line passing through the two points $(3, 1, 0)$ and $(-3, 1, 4)$.

Activity: A plane perpendicular to a line (U2-LO1)

Find the scalar equation of the plane through $P = (4, 2, 1)$ and perpendicular to the line

$$x = 1 + 5t, \quad y = -1 + 4t, \quad z = 2 + 3t.$$

Example: A line of intersection from two planes

We seek to find a parametric equation for the line of intersection of

$$x + y + z = 1, \quad -x + 2y - 3z = -1.$$

First set $z = 0$. Then

$$x + y = 1, \quad -x + 2y = -1.$$

Adding the two equations eliminates x and gives $3y = 1 + (-1) = 0$, so $y = 0$. Substituting into the first equation gives $x = 1 - y = 1$. Thus $P(1, 0, 0)$ is on the line.

Next set $z = 1$. Then

$$x + y = 0, \quad -x + 2y = 2.$$

Adding the two equations eliminates x and gives $3y = 0 + 2 = 2$, so $y = \frac{2}{3}$. Substituting into the first equation gives $x = -y = -\frac{2}{3}$. Thus

$Q(-\frac{2}{3}, \frac{2}{3}, 1)$ is also on the line.

Hence a direction vector is

$$\mathbf{v} = \overrightarrow{PQ} = \begin{bmatrix} -\frac{2}{3} - 1 \\ \frac{2}{3} - 0 \\ 1 - 0 \end{bmatrix} = \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

Thus one parametrization is

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{5}{3} \\ \frac{2}{3} \\ 1 \end{bmatrix}.$$

This example shows how finding the intersection of planes leads naturally to solving systems of linear equations.

Warning: Parallel planes

The method above assumes the two planes intersect in a line. If the normal vectors are parallel, then the planes are either identical or have no intersection.

In Understanding a linear map: a preview, the question “can this output be produced?” became a question about whether a system can be solved.

Activity: Revisiting a requested output as equations (U2-LO1, U2-LO3)

Return to Can this map produce the requested output? (U2-LO1, U2-LO3):

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

Write $A\mathbf{x} = \mathbf{b}$ for the two targets

Activity: Revisiting a requested output as equations (U2-LO1, U2-LO3)
(continued)

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}.$$

What equations do these two targets produce? Which target is reachable?

Matrix form and row operations

A linear system of m equations in n unknowns can be written as one matrix equation

$$A\mathbf{x} = \mathbf{b}.$$

The coefficients form the matrix A , the unknowns form the vector $\mathbf{x}$, and the constants form the vector $\mathbf{b}$. By the definition of the matrix-vector product (Matrix-vector product), the system is consistent exactly when $\mathbf{b}$ is a linear combination of the columns of A .

Activity: Matrix equation for a linear system (U2-LO1)

For

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 6 \\ 2x_1 - 3x_2 + 2x_3 &= 14 \\ 3x_1 + x_2 - x_3 &= -2, \end{aligned}$$

write the system as $A\mathbf{x} = \mathbf{b}$.

Definition: Matrices corresponding to linear equations

The of a linear system of the form the referenced item is the $m \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The is

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

We can adjoin $\mathbf{b}$ to matrix A to create the
representing our linear system:

$$[A \mid \mathbf{b}] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

We will build a method of efficiently solving a large system of linear equations by applying one of several elementary operations to the system, which do not change the solutions to the system, with the hope of eventually simplifying the system to one in which the set of solutions is obvious.

Definition: Elementary operations

The following operations, called ,
can be performed on systems of linear equations, and produce

systems.

1. *Interchanging* the i -th and j -th equations.
2. *Multiplying* an equation by a nonzero constant.
3. *Replacing* the i -th equation by c times the j -th equation *plus* the i -th equation, where $i \neq j$, for some real quantity c .

We can track these operations more easily using an array/matrix to record the coefficients and constants.

Example: Elementary operations with matrices (U2-LO1, U2-LO2)

Consider the linear system

$$\begin{array}{rclcrcl} & y & - & z & = & 4 \\ 2x & & & + & 6z & = & 4 \\ 3x & + & 2y & + & 2z & = & -1 \end{array}$$

This system is reduced to a simpler system using elementary operations. We find the augmented matrix for each linear system and describe the operation used to obtain it from the previous one.

The initial system and augmented matrix are

$$\begin{array}{rcrcrcrcrl} & y & - & z & = & 4 & & \\ 2x & & + & 6z & = & 4 & & \\ 3x & + & 2y & + & 2z & = & -1 & \left[\begin{array}{ccc|c} 0 & 1 & -1 & 4 \\ 2 & 0 & 6 & 4 \\ 3 & 2 & 2 & -1 \end{array} \right] \end{array}$$

Interchange rows 1 and 2:

$$\begin{array}{rcrcrcrcl} 2x & & & + & 6z & = & 4 & \\ & & y & - & z & = & 4 & \\ 3x & + & 2y & + & 2z & = & -1 & \end{array} \quad \left[\begin{array}{ccc|c} 2 & 0 & 6 & 4 \\ 0 & 1 & -1 & 4 \\ 3 & 2 & 2 & -1 \end{array} \right]$$

Divide first row by 2:

$$\begin{array}{rcrcrcrcl} x & & & + & 3z & = & 2 & \\ & & y & - & z & = & 4 & \\ 3x & + & 2y & + & 2z & = & -1 & \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 3 & 2 & 2 & -1 \end{array} \right] \end{array}$$

Replace row 3 by row 3 $-$ 3(row 1):

$$\begin{array}{rcrcrcrl} x & & + & 3z & = & 2 & \\ & y & - & z & = & 4 & \\ & 2y & - & 7z & = & -7 & \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 0 & 2 & -7 & -7 \end{array} \right]$$

Replace row 3 by row 3 $-$ 2(row 2):

$$\begin{array}{rclcrcl} x & + & 3z & = & 2 & & \\ & y & - & z & = & 4 & \\ & & -5z & = & -15 & & \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & -5 & -15 \end{array} \right]$$

Multiply row 3 by $-1/5$:

$$\begin{array}{rclcrcl} x & + & 3z & = & 2 & & \\ & y & - & z & = & 4 & \\ & & z & = & 3 & & \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Replace row 2 by row 2 + row 3:

$$\begin{array}{rclcrcl} x & + & 3z & = & 2 & & \\ & y & & = & 7 & & \\ & & z & = & 3 & & \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 3 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Replace row 1 by row 1 $- 3$ (row 3):

$$\begin{array}{rclcrcl} x & & & = & -7 & & \\ & y & & = & 7 & & \\ & & z & = & 3 & & \end{array} \quad \left[\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Notice that when we apply elementary operations to a system of linear equations, the corresponding rows of their corresponding matrices also change in an analogous way.

Definition: Elementary row operations

An on a matrix M is any one of the following operations:

1. *Interchange* two rows.
2. *Multiply* a row by a nonzero scalar.
3. *Add* a multiple of one row to a different row.

Definition: Row equivalence

Two matrices M and N of the same size are if N can be obtained from M by a finite sequence of elementary row operations.

Warning: Augmented matrices

In the previous two definitions, the matrices M and N can be augmented matrices. In that case, row operations act on every column, including the right-hand-side column.

Theorem

Consider two linear systems each of m equations in n unknowns. If the augmented matrices $[A \mid \mathbf{b}]$ and $[C \mid \mathbf{d}]$ are *row equivalent*, then the linear systems are , i.e., the systems have the same solutions.

The next step is to choose row operations systematically.

Gaussian elimination

Substitution works for small examples. Gaussian elimination is the systematic version: replace a system by an equivalent, simpler system.

Activity: Reading a solution from an augmented matrix (U2-LO1, U2-LO2)

The following augmented matrix (see Matrices corresponding to linear equations) represents a system of linear equations in three unknowns x , y , and z . Write the corresponding equations, then solve the system.

Activity: Reading a solution from an augmented matrix (U2-LO1, U2-LO2)
(continued)

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \end{array} \right]$$

Example: Solving a system using back substitution (U2-LO1, U2-LO2)

We solve the system represented by the following augmented matrix using back substitution.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

We write the corresponding equations:

$$\begin{array}{rcrcrcrcl} x & + & 2y & + & z & = & 4 \\ & & y & + & 2z & = & 2 \\ & & & & z & = & -3 \end{array}$$

Substitute $z = -3$ into the second equation:

$$\begin{array}{rcrcrcrcl} y & + & 2(-3) & = & 2 \\ y & & & = & 8 \end{array}$$

Substitute $y = 8$ and $z = -3$ into the first equation:

$$\begin{array}{rcrcrcrcl} x & + & 2(8) & + & (-3) & = & 4 \\ x & & & & & = & -9 \end{array}$$

Solution: $x = -9$, $y = 8$, and $z = -3$.

Definition: Row echelon form and reduced row echelon form

An $m \times n$ matrix M is in (REF) if it satisfies the following properties:

1. All *zero rows*, if there are any, appear at the *bottom* of the matrix.
2. The *leftmost nonzero entry* in any nonzero row is a 1. This entry is called the of its row.
3. For each nonzero row, the leading one appears to the *right and below* any leading ones in *preceding* rows.

We say M is in (RREF) if M is in REF and also satisfies:

- all columns containing leading ones contain no other non-zero entries.

A matrix in row echelon form appears as a staircase (or *echelon*) pattern of leading ones descending from the upper left corner of the matrix. The figure gives a schematic of a matrix in row echelon form, where the asterisks denote arbitrary constants.

$$\left[\begin{array}{ccccccc} 0 & 1 & * & * & * & * & * \\ 0 & 0 & 0 & 1 & * & * & * \\ 0 & 0 & 0 & 0 & 1 & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Figure. A schematic of a matrix in row echelon form.

Activity: Identifying REF and RREF matrices (U2-LO2)

For each of the following augmented matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$A = \left[\begin{array}{cccc|c} 1 & 3 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Activity: Identifying REF and RREF matrices (U2-LO2)

For each of the following augmented matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$B = \left[\begin{array}{ccc|c} 1 & 4 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 \end{array} \right]$$

Activity: Identifying REF and RREF matrices (U2-LO2)

For each of the following augmented matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$C = \left[\begin{array}{ccc|c} 1 & 3 & 5 & 7 \\ 0 & 1 & -1 & 5 \\ 0 & 0 & 1 & 10 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Activity: Identifying REF and RREF matrices (U2-LO2)

For each of the following augmented matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$D = \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 6 \end{array} \right]$$

Activity: Identifying REF and RREF matrices (U2-LO2)

For each of the following augmented matrices, determine whether it is in row echelon form, reduced row echelon form, both, or neither.

$$E = \left[\begin{array}{ccc|c} 1 & 0 & 4 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & -5 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Definition: Pivot entries, positions, and columns

For a matrix M in REF, each leading one is a . Its location is a . A column containing a pivot position is a .

Theorem

Every matrix M is *row equivalent* to a *unique* matrix in RREF.

The Gaussian Algorithm gives a procedure to find a row-echelon matrix which is row equivalent to any given matrix. The heuristic is to move from top to bottom and outside in until you obtain a matrix in row-echelon form.

Algorithm: The Gaussian Elimination Algorithm

Given a matrix, this algorithm will compute a matrix in row echelon form which is row-equivalent to the original matrix.

1. If the matrix consists entirely of zeros, stop — it is already in row-echelon form.
2. Otherwise, find the left-most column containing a nonzero entry a , and move the row containing that entry to the top position.
3. Now multiply the new top row by $1/a$ so that the row has a leading one.

4. By subtracting multiples of that row from rows below it, make each entry below this leading one equal to zero.

This completes the first row. Now Repeat Steps 1-4 on the matrix consisting of all other rows. The process stops when either no rows remain, or the remaining rows consist entirely of zeros.

The Elimination Game

Activity: Finding row echelon forms (U2-LO2)

Find a row echelon form of the augmented matrix M .

$$M = \left[\begin{array}{ccc|c} 0 & 0 & 1 & 2 \\ 3 & 3 & 6 & -9 \\ 2 & 3 & 0 & -2 \end{array} \right]$$

The Elimination Scratchpad

Solving a linear system

Since we can solve systems of linear equations whose augmented matrices are in row echelon form using back substitution as in Solving a system using back substitution (U2-LO1, U2-LO2), The Gaussian Elimination Algorithm gives a practical method of solving any system of linear equations.

Activity: Solving a linear system using row echelon form (U2-LO2)

Solve the linear system

$$x + 2y + 3z = 9$$

$$2x - y + z = 8$$

$$3x - z = 3$$

by using The Gaussian Elimination Algorithm to find a matrix in row echelon form which is row equivalent to the augmented matrix $[A \mid \mathbf{b}]$.

The Elimination Scratchpad

Note

In Solving a linear system using row echelon form (U2-LO2), every variable column has a pivot. This is why the system has a unique solution.

Definition: Echelon Forms for Linear Systems of Equations

If the augmented matrix of a linear system is a matrix in row echelon form, we say the linear system is in . The variables corresponding to variable columns with leading entries are called (or) , while the other variables are called .

Back substitution allows us to write a general solution to a linear system by writing the basic variables in terms of the free variables.

Example: Solving a system in REF (U2-LO2)

Let

$$[C \mid \mathbf{d}] = \left[\begin{array}{ccccc|c} 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 1 & 2 & 3 & -1 & 7 \\ 0 & 0 & 1 & 2 & 3 & 7 \\ 0 & 0 & 0 & 1 & 2 & 9 \end{array} \right]$$

be the augmented matrix of a linear system. We solve the system using back substitution.

The matrix is already in row echelon form, and so we need only back substitute to solve the basic variables. Let x_1, x_2, x_3, x_4, x_5 be the variables. Then the linear system with the corresponding augmented matrix is given by

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 &= 6 \\ x_2 + 2x_3 + 3x_4 - x_5 &= 7 \\ x_3 + 2x_4 + 3x_5 &= 7 \\ x_4 + 2x_5 &= 9 \end{aligned}$$

The variables x_1, x_2, x_3 , and x_4 are the basic variables, and x_5 is a free variable.

Write the free variable as a parameter t , i.e., write

$$x_5 = t.$$

The last row states that $x_4 + 2x_5 = 9$, which, given that $x_5 = t$, can be rearranged to read that

$$x_4 = 9 - 2t.$$

The third row states that $x_3 + 2x_4 + 3x_5 = 7$. Substituting $x_4 = 9 - 2t$ and $x_5 = t$ into this equation and rearranging, we find that

$$\begin{aligned}
x_3 + 2(9 - 2t) + 3(t) &= 7 \\
x_3 + 18 - 4t + 3t &= 7 \\
x_3 + 18 - t &= 7 \\
x_3 &= -11 + t
\end{aligned}$$

The second row states that $x_2 + 2x_3 + 3x_4 - x_5 = 7$. Substituting $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$ into this equation, and rearranging, we find that

$$\begin{aligned}
x_2 + 2(-11 + t) + 3(9 - 2t) - (t) &= 7 \\
x_2 - 22 + 2t + 27 - 6t - t &= 7 \\
x_2 + 5 - 5t &= 7 \\
x_2 &= 2 + 5t
\end{aligned}$$

Finally, the first row says that $x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 6$. Substituting $x_2 = 2 + 5t$, $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$ into this equation and rearranging, we find that

$$\begin{aligned}
x_1 + 2(2 + 5t) + 3(-11 + t) \\
+ 4(9 - 2t) + 5(t) &= 6 \\
x_1 + 4 + 10t - 33 + 3t \\
+ 36 - 8t + 5t &= 6 \\
x_1 + 7 + 10t &= 6 \\
x_1 &= -1 - 10t
\end{aligned}$$

We now have our solution. All solutions to the linear equation can be written as $x_1 = -1 - 10t$, $x_2 = 2 + 5t$, $x_3 = -11 + t$, $x_4 = 9 - 2t$, and $x_5 = t$, where t is any real number.

Example: Analyzing an inconsistent system (U2-LO2)

Consider a linear system with augmented matrix

$$[C \mid \mathbf{d}] = \left[\begin{array}{cccc|c} 1 & 2 & -3 & 4 & 5 \\ 0 & 1 & 2 & 1 & -6 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right].$$

The last row has the form

$$[0 \ 0 \ 0 \ 0 \mid 1].$$

It represents the equation $0 = 1$, which is impossible. Therefore the system is inconsistent.

Algorithm: Solving $A\mathbf{x} = \mathbf{b}$ by Gaussian elimination

1. Write the augmented matrix $[A \mid \mathbf{b}]$.
2. Row-reduce $[A \mid \mathbf{b}]$ to a row echelon form $[C \mid \mathbf{d}]$.
3. If a row has the form

$$[0 \ 0 \ \cdots \ 0 \mid c], \quad c \neq 0,$$

then the system is inconsistent.

4. Otherwise, pivot columns to the left of the vertical line give basic variables.
5. Non-pivot columns to the left of the vertical line give free variables.
6. Use back substitution to write the basic variables in terms of the free variables.

Note

For an augmented matrix $[A \mid \mathbf{b}]$, only columns to the left of the vertical line are variable columns. A pivot to the right of the vertical line means the system is inconsistent.

Row echelon matrices are not unique for a given starting matrix. For example,

$$\left[\begin{array}{ccc|c} 1 & 3 & 9 & 1 \\ 0 & 1 & -6 & 2 \\ 0 & 0 & 1 & 0 \end{array} \right] \quad \left[\begin{array}{ccc|c} 1 & 3 & 9 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

are row equivalent row echelon matrices, but they are not equal. However, the number of leading ones will be the same in each row equivalent row echelon matrix.

Theorem

Suppose $Ax = b$ is a consistent system with n variables. Let r be the number of pivot columns

in a row echelon form of $[A \mid b]$. Then the solution set is described using exactly $n - r$ parameters.

In particular:

- if $r = n$, the system has a unique solution;
- if $r < n$, the system has infinitely many solutions.

See the theorem. Together with the inconsistency test, it gives a method for finding the number of solutions to any given linear system.

Note: The Number of Solutions to a Linear System

Any linear system belongs to exactly one of three categories.

- *No solution.* This occurs when a row echelon form of $[A \mid b]$ has a row

$$[0 \ 0 \ \cdots \ 0 \mid c], \quad c \neq 0.$$

- *A unique solution.* This occurs when the system is consistent and every variable column, meaning every column to the left of the vertical line, is a pivot column.

- *Infinitely many solutions.* This occurs when the system is consistent and at least one variable column, meaning a column to the left of the vertical line, is not a pivot column.

Note: Row-operation shorthand

We often record row operations using the symbol R_i for row i . The notation $R_i \rightarrow R_i + cR_j$ means to replace row i by row i plus c times row j . All rows not named on the left side stay the same.

- $R_3 \rightarrow R_3 - 2R_1$ means replace row 3 by row 3 minus twice row 1.
- $R_2 \rightarrow \frac{1}{5}R_2$ means multiply row 2 by $1/5$.
- $R_1 \leftrightarrow R_2$ means interchange rows 1 and 2.

Activity: Analyzing solution sets using pivot columns (U2-LO2, U2-LO3)

If

$$C = \left[\begin{array}{ccc|c} 1 & 1 & 2 & 7 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 4 & 11 \end{array} \right]$$

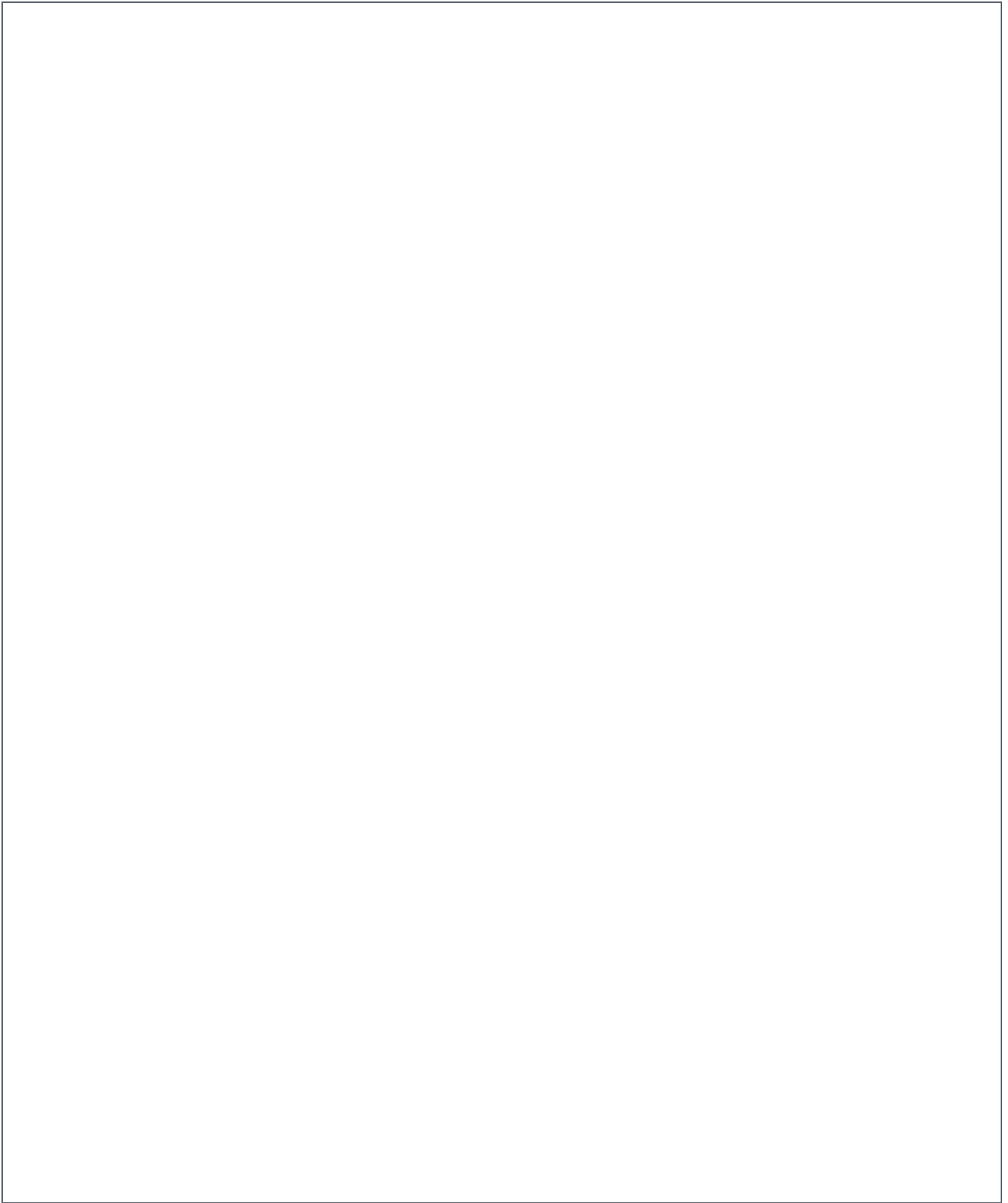
represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?

Activity: Analyzing solution sets using pivot columns (continued) (U2-LO2, U2-LO3)

If

$$D = \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 3 \\ 2 & 1 & 2 & -1 \end{array} \right]$$

represents the augmented matrix for a system of 3 equations and 3 variables, what can we say about the solution set of the system?



Activity: Reading row reduction in code (U2-LO1, U2-LO2)

The following code row reduces the augmented matrix for the two planes

$$x + y + z = 1, \quad -x + 2y - 3z = -1.$$

```
import sympy as sp

M = sp.Matrix([
    [1, 1, 1, 1],
    [-1, 2, -3, -1],
])

M.rref()
```

Output:

```
(Matrix([
[1, 0, 5/3, 1],
[0, 1, -2/3, 0]]), (0, 1))
```

The output has the form

```
(reduced matrix, pivot columns)
```

SymPy numbers columns starting at 0. Thus $(0, 1)$ means the first and second columns.

Read M as an augmented matrix $[A \mid b]$: the first three columns are coefficient columns, and the last column is the right-hand side.

1. Which variable columns are pivot columns?
2. Which variable column is not a pivot column?
3. Let $z = t$. Use the reduced matrix to write x and y in terms of t .

Activity: Reading row reduction in code (U2-LO1, U2-LO2) (continued)

4. Write the solution set in parametric vector form.
5. What kind of solution set is this?
6. How does this connect to the line of intersection of two planes?

Activity: Solving a linear system with NumPy (U2-LO2)

NumPy's `np.linalg.solve(A, b)` computes the solution of $A\mathbf{x} = \mathbf{b}$ when the coefficient matrix A is square and has a pivot in every column. We use the system from Solving a linear system using row echelon form (U2-LO2):

$$x + 2y + 3z = 9,$$

$$2x - y + z = 8,$$

$$3x - z = 3.$$

Enter the coefficient matrix and right-hand side as separate arrays. The command uses floating-point arithmetic.

```
import numpy as np

A = np.array([
    [1, 2, 3],
    [2, -1, 1],
    [3, 0, -1],
], dtype=float)
b = np.array([9, 8, 3], dtype=float)

solution = np.linalg.solve(A, b)
```

Activity: Solving a linear system with NumPy (U2-LO2) (continued)

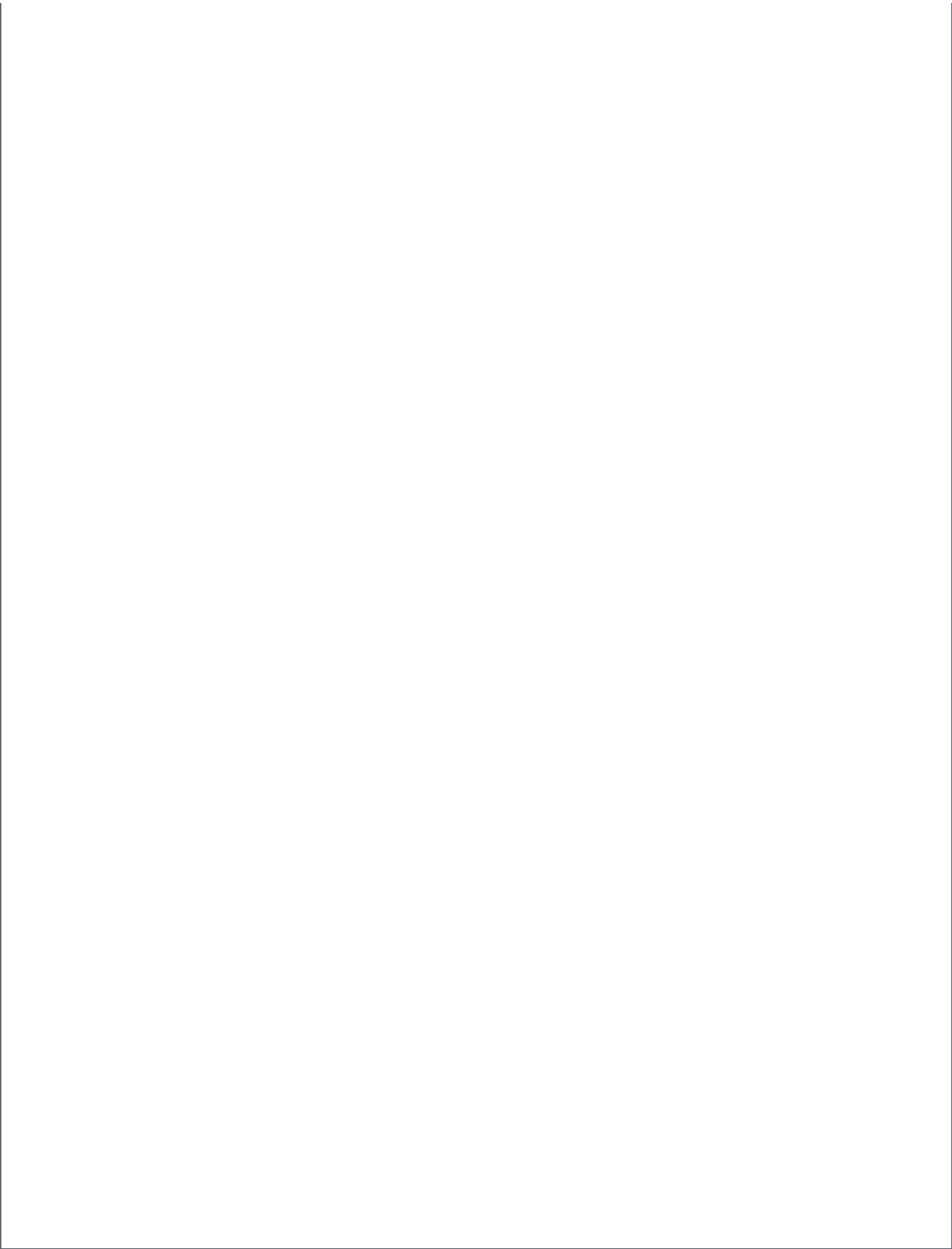
```
print(solution)
print(np.allclose(A @ solution, b))
```

Output:

```
[ 2. -1.  3.]
True
```

The command `np.allclose` checks whether the two arrays agree within a small numerical tolerance.

1. What are the shapes of `A`, `b`, and `solution`?
2. Read the values of x , y , and z from the output. Verify them in the three equations.
3. What does `A @ solution` compute, and what does the output `True` tell us?
4. Why can we not apply `np.linalg.solve` directly to the two-plane system in Reading row reduction in code (U2-LO1, U2-LO2)?



Homogeneous systems and basic solutions

The preview also asked what changes a map forgets. That is the special system $Ax = 0$.

Definition: Homogeneous systems

If a linear system is of the form

$$\begin{array}{ccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & 0 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & 0 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & 0 \end{array}$$

it is called a

- Note that $x_1 = x_2 = \cdots = x_n = 0$ is always a solution to a homogeneous system. It is called the solution.
- A solution to a homogeneous system where *not* all x_i are 0 is called a solution.

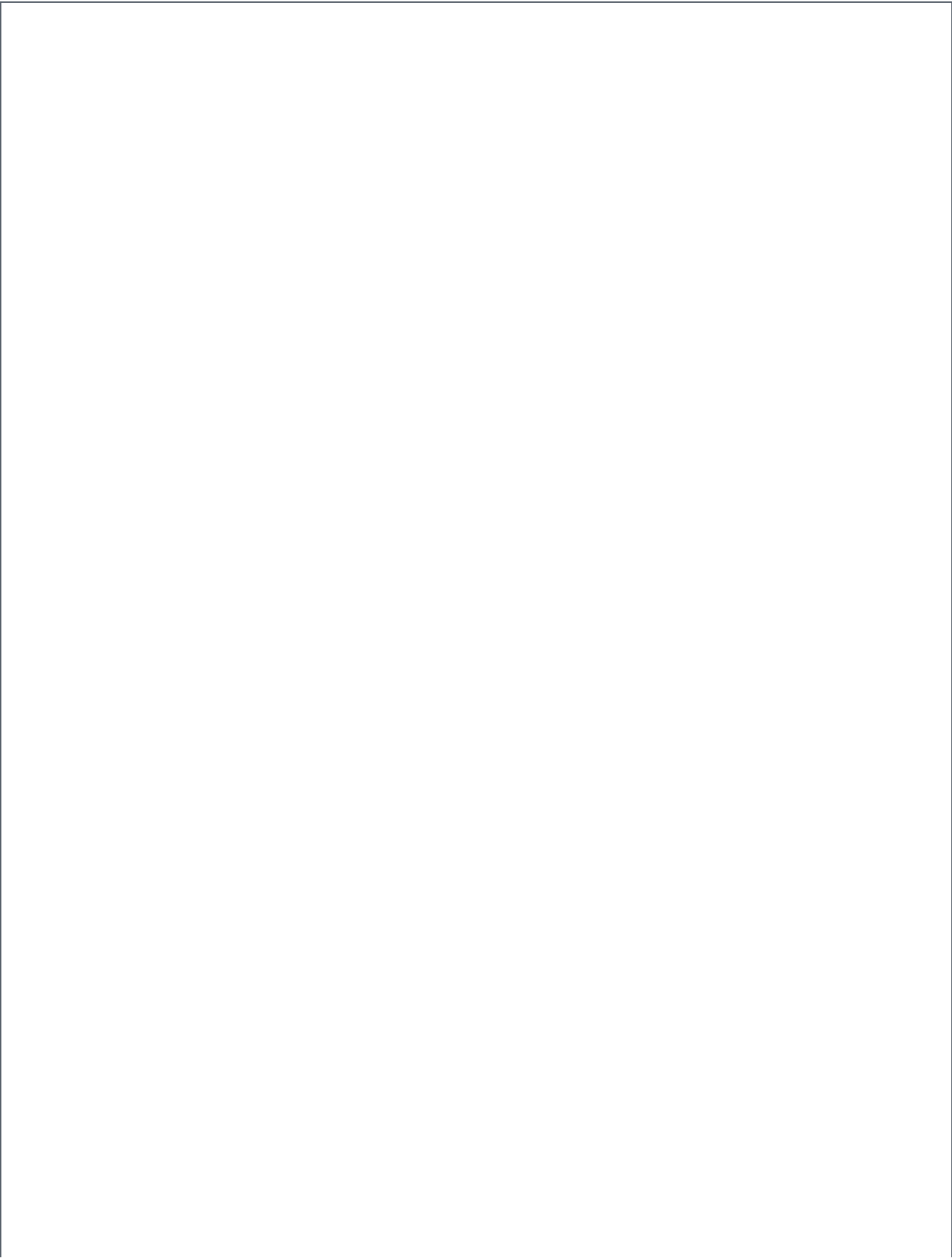
Activity: Solving a homogeneous system (U2-LO2)

Solve the homogeneous system

$$2x_1 + 4x_3 + 6x_4 = 0$$

$$x_1 + 2x_2 + 7x_4 = 0$$

The Elimination Scratchpad



Note: Existence of solutions to homogeneous equations

A homogeneous system of m linear equations in n unknowns *always* has a nontrivial solution if $n > m$, that is if the number of unknowns exceeds the number of equations.

Activity: Thinking about homogeneous systems (U2-L02)

Does a homogeneous system *need* to have more variables than equations in order to have infinitely many solutions?

The solution set of a homogeneous system can be written in vector form.

Activity: Solutions as linear combinations (U2-LO2, U2-LO4)

Rewrite the general solution to Solving a homogeneous system (U2-LO2) as an expression involving a linear combination of certain column vectors.

The Elimination Scratchpad

Fact

If we consider solutions to a homogeneous system as vectors, then any linear combination of solutions to a homogeneous system is again a solution to the homogeneous system.

Activity: Finding basic solutions (U2-LO2, U2-LO4)

Consider the homogeneous system whose augmented matrix, when put in RREF, is

$$[A \mid \mathbf{0}] = \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Solve the system, and write the solutions as a linear combination of particular solutions.

Definition: Basic solutions to homogeneous linear equations

The Gaussian elimination algorithm produces, for each free variable, a solution to the homogeneous equation $A\mathbf{x} = \mathbf{0}$, by setting that free variable equal to one, and all other free variables equal to 0. These solutions

are called , one for every parameter. Every solution can be written uniquely as a linear combination of the basic

solutions.

Theorem

Let A have m rows and n columns, and let r be the number of pivots in a row echelon form of A . Consider the homogeneous system in n variables with A as the coefficient matrix. Then:

1. The system has exactly $n - r$ basic solutions, one for each parameter. Note that $n - r$ is also the number of *free* variables of the linear system when the augmented matrix $[A \mid \mathbf{0}]$ is reduced to row echelon form (recall Echelon Forms for Linear Systems of Equations).
2. Every solution is a linear combination of these basic solutions.

Activity: A plane through three points without cross products (U2-LO1, U2-LO2)

Use Gaussian elimination and basic solutions to find an equation for the plane containing

$$P(1, 1, -2), \quad Q(0, 2, 1), \quad R(-1, -1, 0),$$

(*without* using the cross-product, if you happen to know what that is).

For a homogeneous system, row-reducing $[A \mid 0]$ is equivalent to row-reducing A while remembering that the right-hand side stays zero. The next section gives the solution set of $Ax = 0$ a name, and from that point on we work with A alone.

Homogeneous solution sets are closed under linear combinations. Subspaces gives a name to sets with this behavior: subspaces.